

1. MORPHOLOGICAL ANALYSIS PIPELINE

```
PS C:\COURSE\DSA0302\CO2_AT1> & "C:\Users\Shivaji V\AppData\Local\Microsoft\Windows\Apps\Windows Defender Security Center\NLP 1.py"
```

Word	Root	Suffix	Type	Normalized
connected	connect	ed	Inflectional	connect
connecting	connect	ing	Inflectional	connect
connection	connect	ion	Derivational	connect

2. MORPHOLOGICAL PARSING MODULE

```
PS C:\COURSE\DSA0302\CO2_AT1> & "C:\Users\Shivaji V\AppData\Local\Microsoft\Windows\Apps\Windows Defender Security Center\NLP 2.py"
```

Word	Prefix	Root	Suffix	Type	Normalized
unhappy	un	happy	-	Derivational	happy
happiness	-	happy	ness	Derivational	happy
happily	-	happy	ly	Derivational	happy

```
PS C:\COURSE\DSA0302\CO2_AT1>
```

3. STEMMING BASED PREPROCESSING MODULE

```
PS C:\COURSE\DSA0302\CO2_AT1> & "C:\Users\Shivaji V\AppData\Local\Microsoft\Windows\Apps\Windows Defender Security Center\NLP 3.py"
```

Word	Stem	Affix	Type	Normalized
played	play	ed	Inflectional	play
player	play	er	Derivational	play
playing	play	ing	Inflectional	play

```
PS C:\COURSE\DSA0302\CO2_AT1>
```

4. FINITE STATE MORPHOLOGICAL PARSER

```
PS C:\COURSE\DSA0302\CO2_AT1> & "C:\Users\Shivaji V\AppData\Local\Microsoft\Windows\Apps\Windows Defender Security Center\NLP 4.py"
```

Word	State Path	Pattern	Root	Normalized
writes	Start->Verb->+s->End	Regular	write	write
writing	Start->Verb->+ing->End	Regular	write	write
written	Start->Irregular->End	Irregular	write	write

5. PORTER STEMMER BASED PREPROCESSING

```
PS C:\COURSE\DSA0302\CO2_AT1> & "C:\Users\Shivaji V\AppData\Local\Microsoft\Windows\Apps\Windows Defender Security Center\NLP 5.py"
```

Word	Intermediate	Final Stem
relational	relation	relat
relation	relat	relat
relate	rel	relat